AHMAD ESSAMELDIN AHMAD

Mechanical Design Engineer

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https://www.linkedin.com/in/ahmad-esaameldin-ahmad-b266321b3/ Saint Paul, Minnesota, USA (U.S. Citizen)

SUMMARY

I am Ahmad, a U.S. citizen born in Minnesota, currently working in a hybrid role from Egypt as a Mechanical Design Engineer. I'm planning a permanent relocation to the U.S. and actively seeking opportunities in robotics and industrial automation. I specialize in developing innovative mechanical solutions tailored to industry-specific challenges, with a strong focus on robotics integration, advanced mechanical design, and complex surface applications. I'm committed to advancing my career by continuously improving both my technical and interpersonal skills.

EXPERIENCE

08/2023 - Present
Cairo, Egypt

Mechanical Design Engineer (Hybrid)

ARIA Technologies

Engineering firm specializing in innovative technology solutions

- Design and develop mechanical systems and assemblies for robotic automation projects, including the Aria Robot P60.
- Create detailed 2D and 3D engineering drawings using SolidWorks and AutoCAD for manufacturing and prototyping.
- Lead the transition of component manufacturing to in-house CNC machining, reducing lead time and outsourcing costs by 30%.
- Collaborate with cross-functional teams in R&D, electrical, and production departments to ensure full system integration and performance optimization.
- Participate in DFMA (Design for Manufacturing and Assembly) reviews to improve design efficiency and manufacturability.
- Implement revision control and maintain drawing standards per ISO/ASME standards.
- Utilized advanced software tools and best practices for high-quality solutions.
- Increased design efficiency by 25% through implementing new CAD software tools, resulting in a $100k cost reduction.
- Improved project delivery time by 15% by streamlining the design phase workflow, saving $200k annually.
- Enhanced product lifespan by 40% by integrating advanced FEA simulations into design process for critical components.

EDUCATION

09/2018 - 07/2023
Giza, Egypt

B.S. in Mechatronics Engineering

MSA University

Second Ranked GPA: **3.94** / **4.00**

09/2015 - 07/2018
Ahsaa, KSA

High School Diploma

Dor - Al uloom Schools

LANGUAGES

Arabic
Native

English
Proficient

STRENGTHS

Leadership and Time Management
Strong leadership and time management capabilities

Presentation Skills
Excellent presentation skills with the ability to convey complex information

Cross-functional collaboration
Effective in collaboration and design reviews.

Innovative Problem Solver
Skilled in developing innovative mechanical solutions for industry challenges.

Advanced CAD Proficiency
Proficient in creating detailed engineering drawings using SolidWorks.

Cost Reduction Expertise
Experienced in reducing manufacturing costs through in-house machining.

Collaborative Team Leader
Effective at integrating cross-functional teams for project success.

PROJECTS

Flight Simulator Moving Platform and Active Controls (Part 2)

02/2023 - 06/2023 Giza, Egypt
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Graduation Project Part 2 focusing on flight simulation

- Led the development of a 6-DOF motion simulator rig to replicate realistic pilot feedback. Included development of active joystick/yoke and pedal peripherals for full immersion.
- Oversaw full mechanical design, modeling, and control integration.
- Applied control systems using Arduino and inverse kinematics for platform motion.

Flight Simulator Fixed Platform and Pilot Controls (Part 1)

08/2022 - 01/2023 Giza, Egypt
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Graduation Project Part 1 emphasizing fixed simulation technology

- Improved training efficiency by {25%} by designing and building a simulator cockpit with custom input peripherals.
- Enhanced mechanical ergonomics, improving system efficiency by {15%} through innovative control system design.
- Focused on user-centered ergonomics and realistic control positioning.
- Established modular design for later upgrade into a motion platform.

EMG- Based Prosthetic Arm with Force Feedback (Gen.2)

08/2022 - 01/2023 Giza, Egypt
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Led development of a 3D-printed smart prosthetic arm using EMG control and force feedback integration

- Achieved a {30%} cost reduction in prosthetic arms by designing one with EMG signal processing technologies.
- Implemented feedback loop using vibration motors for sensory output.
- Integrated 3D-printed components with servo-actuated fingers.
- Presented at Innovation Catalyst, focusing on accessibility and cost-efficiency.

AI Real-Time Hand Gesture Prosthetic Arm (Gen.1)

01/2022 - 04/2022 Giza, Egypt
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Led a team to create a low-cost, lightweight prosthetic arm controlled by real-time hand gestures.

- Used computer vision and glove-based sensors for gesture recognition.
- Developed real-time control logic to actuate fingers based on gestures.
- Manufactured all parts using rapid prototyping methods.
- Targeted for users with upper-limb disability in developing regions.

Design and Manufacturing of a 3D Printer

01/2021 - 03/2021 Giza, Egypt
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Led the complete design and construction of a custom 3D printer for academic prototyping use.

- Focused on FDM principles, mechanical stability, and cost-efficiency.
- Designed Cartesian-style 3D printer frame and motion system.
- Selected and installed stepper motors, drivers, and extruder system.
- Programmed firmware (Marlin) and calibrated machine manually.

KEY ACHIEVEMENTS

Honor Shield Winner

Winning the first place for my graduation project at the 16th UGRF hosted by Nile University and the Egyptian Knowledge Bank

PUBLICATIONS

Designing an Optimal Washout Filter for a Flight Simulator Using Linear Quadratic Regulator (LQR)

Arabian Journal for Science and Engineering

06/2023

Flight Simulator Motion Cueing Platform: Interfacing and Pilot Controls Prototyping

MSA Engineering Journal

02/2023

SKILLS

Technical Skills

Animation, Autodesk Fusion 360, FEA, IAM, Inventor, PVSyst, Robotics, SolidWorks, CAD.